# LUISS

Academic Year 2024/2025

*Mathematical Methods for Finance*

## Linear algebra, linear applications, change of basis.

### 1. Linear applications

**Exercise 1.** *Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be the application defined by*

$$T(x, y, z) = (x - y, x - 3y + 2z)\,.$$

*Prove that $T$ is a linear map, and find the matrix $A$ such that*

$$T(x, y, z) = A \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad \text{for all } (x, y, z) \in \mathbb{R}^3\,.$$

**Solutions.** We have to check that, for all $\mathbf{x}$, $\mathbf{x_1}$, $\mathbf{x_2} \in \mathbb{R}^3$, and for all $\lambda \in \mathbb{R}$, it holds

- $T(\mathbf{x_1} + \mathbf{x_2}) = T(\mathbf{x_1}) + T(\mathbf{x_2})$;
- $T(\lambda \mathbf{x}) = \lambda T(\mathbf{x})$.

For the first point, let $\mathbf{x_1} = (x_1, y_1, z_1)$, $\mathbf{x_2} = (x_2, y_2, z_2)$. Then we have

$$\begin{aligned} T(\mathbf{x_1} + \mathbf{x_2}) &= T(x_1 + x_2, y_1 + y_2, z_1 + z_2) \\ &= (x_1 + x_2 - (y_1 + y_2), x_1 + x_2 - 3(y_1 + y_2) + 2(z_1 + z_2)) \\ &= \big((x_1 - y_1) + (x_2 - y_2), (x_1 - 3y_1 + 2z_1) + (x_2 - 3y_2 + 2z_2)\big) \\ &= (x_1 - y_1, x_1 - 3y_1 + 2z_1) + (x_2 - y_2, x_2 - 3y_2 + 2z_2) = T(\mathbf{x_1}) + T(\mathbf{x_2})\,. \end{aligned}$$

In the same way, for $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$ and $\lambda \in \mathbb{R}$,

$$\begin{aligned} T(\lambda \mathbf{x}) &= T(\lambda x, \lambda y, \lambda z) \\ &= (\lambda x + \lambda y, \lambda x - 3\lambda y + 2\lambda z) \\ &= \big(\lambda(x + y), \lambda(x - 3y + 2z)\big) = \lambda(x + y, x - 3y + 2z) = \lambda T(\mathbf{x})\,. \end{aligned}$$

Then $T$ is linear, and so there exists a matrix $A$ such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^3$.

Since $\mathbf{x} \in \mathbb{R}^3$ and $A\mathbf{x} \in \mathbb{R}^2$, $A$ must be a $2 \times 3$ matrix (2 rows and 3 columns). This means

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}, \qquad \text{for some } a_{ij} \in \mathbb{R}\,, \quad 1 \le i \le 2\,, \quad 1 \le j \le 3\,.$$

To find $A$ we compute $A\mathbf{x}$, for $\mathbf{x} = (x, y, z) \in \mathbb{R}^3$. We have

$$A\mathbf{x} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y + a_{13}z \\ a_{21}x + a_{22}y + a_{23}z \end{pmatrix}$$

Since we must have $T(\mathbf{x}) = A\mathbf{x}$, this implies

$$\begin{pmatrix} a_{11}x + a_{12}y + a_{13}z \\ a_{21}x + a_{22}y + a_{23}z \end{pmatrix} = \begin{pmatrix} x - y \\ x - 3y + z \end{pmatrix} \implies \begin{array}{lll} a_{11} = 1\,, & a_{12} = -1\,, & a_{13} = 0 \\ a_{21} = 1\,, & a_{22} = -3\,, & a_{23} = 1\,. \end{array}$$

This means

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -3 & 1 \end{pmatrix}, \qquad T(x,y,z) = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

**Remark 1.**

- The linearity of $T$ can also be proved in the following way: $T$ is linear if and only if, for all $\mathbf{x_1}$, $\mathbf{x_2} \in \mathbb{R}^3$, and for all $\lambda, \mu \in \mathbb{R}$, the following equality holds
$$T(\lambda \mathbf{x_1} + \mu \mathbf{x_2}) = \lambda T(\mathbf{x_1}) + \mu T(\mathbf{x_2})\,;$$
- We also know that $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if there exists a matrix $A \in Mat(m \times n)$ such that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Hence, if we directly find such matrix we can conclude that $T$ must be linear;
- Observe in the previous exercise that the first (resp. second) row of $A$ represents the coefficients of $x, y, z$ in the first (resp. second) equation. This is a general fact and can be used in the exercises to find quickly the coefficients of the matrix $A$;
- Consider the *canonical basis* $\mathscr{B} = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, where $\boldsymbol{e}_1 = (1,0,0)$, $\boldsymbol{e}_2 = (0,1,0)$, $\boldsymbol{e}_3 = (0,0,1)$. Then the columns of $A$ represent the images of the canonical basis through the application $T$, i.e. the first column is given by $T(\boldsymbol{e}_1)$, the second one by $T(\boldsymbol{e}_2)$, the third one by $T(\boldsymbol{e}_3)$.

**Exercise 2.** *Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be the application defined as*

$$T(x,y) = (x + y + 1,\, 2x,\, 3x - y + 2)\,.$$

*Prove that $T$ is <u>not</u> a linear map. Prove, moreover, that there exist a matrix $A \in Mat(3 \times 2)$ and a vector $\boldsymbol{b} \in \mathbb{R}^3$ such that $T(\boldsymbol{x}) = A\boldsymbol{x} + \boldsymbol{b}$.*

**Solution.** To prove that $T$ is not linear, we have to prove that at least one of the equalities $T(\boldsymbol{x}_1 + \boldsymbol{x}_2) = T(\boldsymbol{x}_1) + T(\boldsymbol{x}_2)$, $T(\lambda \boldsymbol{x}) = \lambda T(\boldsymbol{x})$, is false.

We show a counterexample for the second one (of course there are counterexamples also for the first one). Observe that we have, for $\boldsymbol{x} = (x, y) \in \mathbb{R}^2$,

$$T(\lambda \boldsymbol{x}) = T(\lambda x, \lambda y) = (\lambda x + \lambda y + 1,\, 2\lambda x,\, 2\lambda x - \lambda y + 2)\,,$$
$$\lambda T(\boldsymbol{x}) = \lambda T(x,y) = \lambda(x + y + 1,\, 2x,\, 2x - y + 2) = (\lambda x + \lambda y + \lambda,\, 2\lambda x,\, 2\lambda x - \lambda y + 2\lambda)\,.$$

We note immediately that the two quantities are different. For example, for $\boldsymbol{x} = (0,0)$ and $\lambda = 0$, we have $\lambda T(\boldsymbol{x}) = (0,0,0)$, whereas $T(\lambda \boldsymbol{x}) = T(0,0) = (1,0,2)$.

To find the matrix $A$ and the vector $\boldsymbol{b}$, we impose the condition $T(\boldsymbol{x}) = A\boldsymbol{x} + \boldsymbol{b}$. Since $A \in Mat(3 \times 2)$ and $\boldsymbol{b} \in \mathbb{R}^3$, we have

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

Hence

$$A\boldsymbol{x} + \boldsymbol{b} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y + b_1 \\ a_{21}x + a_{22}y + b_2 \\ a_{31}x + a_{32}y + b_3 \end{pmatrix}$$

The condition $T(\boldsymbol{x}) = A\boldsymbol{x} + \boldsymbol{b}$ implies

$$\begin{pmatrix} x+y+1 \\ 2x \\ 3x-y+2 \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y + b_1 \\ a_{21}x + a_{22}y + b_2 \\ a_{31}x + a_{32}y + b_3 \end{pmatrix} \implies \begin{array}{lll} a_{11} = 1\,, & a_{12} = 1\,, & b_1 = 1 \\ a_{21} = 2\,, & a_{22} = 0\,, & b_2 = 0\,, \\ a_{31} = 3\,, & a_{32} = -1\,, & b_3 = 2\,. \end{array}$$

This means

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 0 \\ 3 & -1 \end{pmatrix}\,, \qquad \boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}\,, \qquad T(x,y,z) = \begin{pmatrix} 1 & 1 \\ 2 & 0 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}\,.$$

**Remark 2.**

- An application $T(\boldsymbol{x}) = A\boldsymbol{x}+\boldsymbol{b}$ is called *affine*. It can be written as a sum of two applications $T = T_1 + T_2$, where $T_1$ is the linear map $T_1(\boldsymbol{x}) = A\boldsymbol{x}$, whereas $T_2$ is the constant map $T_2(\boldsymbol{x}) = \boldsymbol{b}$. Affine maps can be seen as translation of a linear maps;
- For $n = 1$, affine maps $T(x) = ax + b$ are straight lines, whereas linear maps $T(x) = ax$ are straight lines passing from the origin;
- As before, the rows of $A$ represents the coefficients of $x, y$ in the equations composing $T(\boldsymbol{x})$, whereas $\boldsymbol{b}$ is the vector composed by the known terms of the equations.

## 2. Change of basis

We start by recalling some basic facts about linear algebra.

2.1. **Linear dependence and independence.** A set of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space $V$ is said to be *linearly independent* if no vector in the set can be expressed as a linear combination of the others.

Mathematically, the set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is linearly independent if the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n = \mathbf{0}$$

has only the trivial solution $c_1 = c_2 = \ldots = c_n = 0$.

Conversely, a set of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is said to be *linearly dependent* if there exist coefficients $c_1, c_2, \ldots, c_n$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n = \mathbf{0}.$$

In this case, at least one vector in the set can be expressed as a linear combination of the others, leading to a non-trivial solution to the equation.

For example, the vectors $\boldsymbol{v}_1 = (2, 1)$ and $\boldsymbol{v}_2 = (1, -1)$ are linearly independent. We can prove this writing the equation

$$\alpha\boldsymbol{v}_1 + \beta\boldsymbol{v}_2 = \mathbf{0} \implies (2\alpha + \beta, \alpha - \beta) = (0, 0) \implies \begin{cases} 2\alpha + \beta = 0\,, \\ \alpha - \beta = 0\,. \end{cases}$$

From the second equation we get $\alpha = \beta$. Substituting into the first one, we get $3\beta = 0$, which implies $\alpha = \beta = 0$ and proves the linear independence.

Conversely, the vectors $\boldsymbol{v}_1 = (1, 3)$, $\boldsymbol{v}_2 = (2, 6)$ are linearly dependent. The system becomes here

$$\alpha\boldsymbol{v}_1 + \beta\boldsymbol{v}_2 = 0 \implies \begin{cases} \alpha + 2\beta = 0\,, \\ 2\alpha + 6\beta = 0\,. \end{cases}$$

From the first equation we get $\alpha = -2\beta$. Substituting into the second one we have $-6\beta + 6\beta = 0$, which admits infinite solution. Hence, choosing for example $\beta = -1$ and $\alpha = 2$ we have

$$2\boldsymbol{v}_1 - \boldsymbol{v}_2 = 0\,,$$

which proves the linear dependence.

Geometrically speaking, two vectors are linearly dependent if they are parallel (one of them is a multiple of the other one). Three vectors are linearly dependent if they belongs to the same plane.

2.2. **Rank of a matrix.** Given a matrix $A \in Mat(n \times m)$, the *rank* of $A$, denoted by $\mathrm{rk}(A)$, is the maximum number of rows (or columns) linearly independent.

We always have $\mathrm{rk}(A) \leq \min\{n, m\}$. It can be proved that $\mathrm{rk}(A) = k$ if:

- There exists a $k \times k$ submatrix of $A$ with non-zero determinant;
- all the square submatrices of order higher than $k$ have zero determinant.

For example, the matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 3 & 2 & 1 \end{pmatrix}$$

has rank 2. To prove that, we have to check that there exist a $2 \times 2$ submatrix of $A$ with non-zero determinant, and that the matrix $A$ (which is the only square submatrix with order $> 2$) has zero determinant. We have

$$\det A = \det \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 3 & 2 & 1 \end{pmatrix} = 0\,, \qquad \det \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} = -4 \neq 0\,.$$

Of course, this gives us a powerful method to check if a set of vectors is linearly independent. We have that $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ is linearly independent if the matrix having $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ as rows (or columns) has rank $n$.

2.3. **Basis and dimension.** A *basis* for a vector space (or subspace) $V$ is a set of linearly independent vectors in $V$ that spans $V$, meaning that any vector in $V$ can be expressed as a unique linear combination of the basis vectors. Formally, a set of vectors $\mathscr{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a basis for $V$ if:

1. The vectors are linearly independent;
2. The vectors span $V$, i.e., for any vector $\mathbf{v}$ in $V$, there exist scalars $c_1, c_2, \ldots, c_n$ such that

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n.$$

If $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for $V$, then the number $n$ is called the *dimension* of $V$, and we write $\dim(V) = n$. In other words, the dimension of $V$ is defined as the maximum number of linearly independent vectors that can be chosen from $V$.

When $V = \mathbb{R}^n$, the most common basis is given by the *canonical basis* $\mathscr{E} = \{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$, where

$$\boldsymbol{e}_1 = (1, 0, \ldots, 0)\,, \quad \boldsymbol{e}_2 = (0, 1, 0, \ldots, 0)\,, \quad \ldots \quad \boldsymbol{e}_n = (0, \ldots, 0, 1)\,.$$

In general, $(\boldsymbol{e}_i)_i = 1$ and $(\boldsymbol{e}_i)_j = 0$ if $j \neq i$.

Observe that, when we write the coordinates of a vector, we are always working with the canonical basis. For example, writing $\boldsymbol{v} = (2, 4, 3, -2)$ in $\mathbb{R}^4$ means

$$\boldsymbol{v} = (2, 4, 1, -1) = 2\boldsymbol{e}_1 + 4\boldsymbol{e}_2 + 3\boldsymbol{e}_3 - 2\boldsymbol{e}_4\,.$$

2.4. **Change of basis.** We are not forced to work with the canonical basis. For example, consider the set $\mathscr{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ in $\mathbb{R}^3$, where $\boldsymbol{v}_1 = (1,1,0)$, $\boldsymbol{v}_2 = (0,1,-1)$, $\boldsymbol{v}_3 = (1,1,1)$.

We want to prove that $\mathscr{B}$ is a basis for $\mathbb{R}^3$. Since $\dim(\mathbb{R}^3) = 3$, it suffices to prove that the vectors are linearly independent, i.e. the matrix formed by $\boldsymbol{v}_1$, $\boldsymbol{v}_2$, $\boldsymbol{v}_3$ has non-zero determinant. We have

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix} = 1 \neq 0 \implies \mathscr{B} \text{ is a basis for } \mathbb{R}^3 .$$

Then we can write every vector in $\mathbb{R}^3$ as a linear combination of the vectors of $\mathscr{B}$. For example, the vector $\boldsymbol{v} = (3,4,0)$ can be written as

$$\boldsymbol{v} = (1,2,-2) = 2(1,1,0) + (0,1,-1) - (1,1,1) = 2\boldsymbol{v}_1 + \boldsymbol{v}_2 - \boldsymbol{v}_3 .$$

We write $\boldsymbol{v} = (2,1,-1)_{\mathscr{B}}$. Of course we have $(1,0,0)_{\mathscr{B}} = \boldsymbol{v}_1$, $(0,1,0)_{\mathscr{B}} = \boldsymbol{v}_2$, $(0,0,1)_{\mathscr{B}} = \boldsymbol{v}_3$.

From here, two natural questions arise:

- How can we build a map $\Phi$ that takes a vector in the basis $\mathscr{B}$ and writes it in the canonical basis $\mathscr{E}$?
- How can we build a map $\Psi$ that takes a vector in the canonical basis $\mathscr{E}$ and writes it in the basis $\mathscr{B}$?

Observe that both maps are linear. Then we just have to find the matrices describing the two applications.

We call $B$ the matrix describing the maps $\Phi$. For $\boldsymbol{v} = (x,y,z)_{\mathscr{B}}$, we have

$$\Phi(\boldsymbol{v}) = x\boldsymbol{v}_1 + y\boldsymbol{v}_2 + z\boldsymbol{v}_3 = x(1,1,0) + y(0,1,-1) + z(1,1,1) = (x+z, x+y+z, -y+z) .$$

Thus, $\Phi(\boldsymbol{v}) = B\boldsymbol{v}$, with

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} .$$

Observe that the columns of $B$, as said in Remark 1, are given by $\boldsymbol{v}_1 = \Phi(1,0,0)$, $\boldsymbol{v}_2 = \Phi(0,1,0)$, $\boldsymbol{v}_3 = \Phi(0,0,1)$.

As regards the function $\Psi$, observe that $\Psi = \Phi^{-1}$. Hence, the matrix describing $\Psi$ is given by the inverse matrix of $B$, $\Psi(\boldsymbol{v}) = B^{-1}\boldsymbol{v}$.

2.5. **Change of basis in linear applications.** Let $T(\boldsymbol{v}) = A\boldsymbol{v}$ a linear application. The matrix $A$ describing $T$, if nothing else is specified, is always referred to the canonical basis $\mathscr{E}$. Hence, if $\boldsymbol{v}$ is written in the basis $\mathscr{E}$, the image $T(\boldsymbol{v})$, written in the canonical basis $\mathscr{E}$, is given by $A\boldsymbol{v}$.

*Problem:* Given another basis $\mathscr{B}$, what is the matrix describing $T$ in the new basis $\mathscr{B}$?

Hence, taken a vector $\boldsymbol{v}$ written in the basis $\mathscr{B}$, how can we write $T(\boldsymbol{v})$ in the new basis $\mathscr{B}$?

We can split this problem into three steps:

- First of all, we take $\boldsymbol{v}$ and we convert it in the canonical basis $\mathscr{E}$. As said before, the new vector is $B\boldsymbol{v}$, where the columns of $B$ are the vectors of the basis $\mathscr{B}$;
- Then, we compute $T(B\boldsymbol{v})$ in the basis $\mathscr{E}$. Since we are working with the canonical basis, this image will be $A(B\boldsymbol{v}) = AB\boldsymbol{v}$;
- Then, we convert $AB\boldsymbol{v}$, which is written now in the basis $\mathscr{E}$, into the basis $\mathscr{B}$. As said before, the new vector is $B^{-1}(AB\boldsymbol{v}) = (B^{-1}AB)\boldsymbol{v}$.

Hence, *the linear map $T$, with respect to the basis $\mathscr{B}$, is described by the matrix $B^{-1}AB$.*

2.6. **Inverse of a matrix.** To conclude, we recall the formula for the inverse of a matrix. As we saw in the previous step, the inverse of a matrix plays a crucial role to compute the matrix describing a linear application in a basis different than the canonical one.

Let $A$ be a $n \times n$ square matrix, with $\det(A) \neq 0$. To compute the inverse $A^{-1}$, we need to define the so-called *cofactor matrix*.

The cofactor matrix, denoted as $\mathrm{Cof}(A)$, us a matrix in which each element is the cofactor of the corresponding element in matrix $A$. The cofactor of an element $a_{ij}$ in $A$ is denoted as $\mathrm{Cofactor}(a_{ij})$ and is calculated as follows:

$$\mathrm{Cofactor}(a_{ij}) = (-1)^{i+j} \cdot M_{ij},$$

where $M_{ij}$ is the determinant of the matrix obtained by deleting the $i$-th row and $j$-th column from $A$.

The inverse of a matrix $A$ can be computed as follows:

$$A^{-1} = \frac{1}{\det A} \mathrm{Cof}(A)^T,$$

where $\mathrm{Cof}(A)^T$ denotes the transpose of $\mathrm{Cof}(A)$.

For example, we consider a $3 \times 3$ matrix $A$:

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

To find the inverse of matrix $A$, we start computing the determinant, $\det(A) = -4$. Since $\det(A) \neq 0$, the matrix $A$ is invertible.

Now we compute the cofactor matrix $\mathrm{Cof}(A)$. Just to give an example, we report the cofactor of the element $a_{12} = 1$:

$$\mathrm{Cofactor}(a_{12}) = (-1)^{1+2} \cdot \det \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix} = -4$$

The cofactor matrix is the following one (check the computations):

$$\mathrm{Cof}(A) = \begin{pmatrix} 2 & -4 & -4 \\ -1 & 0 & 2 \\ -2 & 4 & 0 \end{pmatrix}$$

Then we calculate the inverse of $A$ by using the formula:

$$A^{-1} = \frac{1}{\det(A)} \mathrm{Cof}(A)^T = -\frac{1}{4} \cdot \begin{pmatrix} 2 & -1 & -2 \\ -4 & 0 & 4 \\ -4 & 2 & 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{4} & \frac{1}{2} \\ 1 & 0 & -1 \\ 1 & -\frac{1}{2} & 0 \end{pmatrix}$$

**Exercise 3.** *Consider the linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by $T(x, y) = (x - y, 2x + y)$.*

- *Find the matrix $A$ describing $T$;*
- *Prove that $\mathscr{B} = \{\boldsymbol{e}_1 + \boldsymbol{e}_2, 2\boldsymbol{e}_1\}$ is a basis of $\mathbb{R}^2$;*
- *Find the matrix describing $T$ with the respect of the basis $\mathscr{B}$.*

**Solution:** The matrix $A$ can be immediately computed:

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}.$$

The set $\mathscr{B}$ is given by $\mathscr{B} = \{(1,1),(2,0)\}$. Since the determinant

$$\det \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} = -2\,,$$

is different from 0, then $\mathscr{B}$ is a basis for $\mathbb{R}^2$.

As said before, the matrix describing $T$ with respect to $\mathscr{B}$ is $B^{-1}AB$, where

$$B = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$$

is composed by the vectors of $\mathscr{B}$ as columns. Then we compute $B^{-1}$. Recall that the inverse of a $2 \times 2$ matrix can be computed in the following way

$$C = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \implies C^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Then we have

$$B^{-1} = -\frac{1}{2} \begin{pmatrix} 0 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

Then we can conclude:

$$B^{-1}AB = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ -\frac{3}{2} & -1 \end{pmatrix}.$$

**Exercise 4.** *Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear map defined as*

$$T(x,y,z) = (x+z, y+z, 3x+5y+8z)\,.$$

1. *Find the matrix associated with $T$ with respect to the canonical basis;*
2. *Find the matrix associated with $T$ with respect to the basis $\mathscr{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$, where $\boldsymbol{v}_1 = \boldsymbol{e}_1 + \boldsymbol{e}_2 + \boldsymbol{e}_3$, $\boldsymbol{v}_2 = \boldsymbol{e}_3$ and $\boldsymbol{v}_3 = 3\boldsymbol{e}_2 + 3\boldsymbol{e}_3$;*
3. *Determine, with respect to the canonical basis, the dimensions and the basis of $Ker(T)$ and $Im(T)$;*
4. *Say if the vectors $\boldsymbol{v} = 2\boldsymbol{e}_1 + 5\boldsymbol{e}_2$ and $w = 6\boldsymbol{v}_1 + 24\boldsymbol{v}_2 - \boldsymbol{v}_3$ belong to $Im(T)$. Finally, say if $\boldsymbol{z} = 4\boldsymbol{e}_1 + 2\boldsymbol{e}_2 - 2\boldsymbol{e}_3$ belongs to $Ker(T)$.*

**Solution.** *Point 1.* As in the previous exercise, we can compute quickly the matrix $A$, recalling that the rows are composed by the coefficients of $x, y, z$ in the three components of $T(x,y,z)$.

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 3 & 5 & 8 \end{pmatrix}$$

*Point 2.* The set $\mathscr{B}$ is given by $\mathscr{B} = \{(1,1,1),(0,0,1),(0,3,3)\}$. Since the determinant

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 3 & 3 \end{pmatrix} = -3\,,$$

is different from 0, then $\mathscr{B}$ is a basis for $\mathbb{R}^3$.

As said before, the matrix describing $T$ with respect to $\mathscr{B}$ is $B^{-1}AB$, where

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 3 \\ 1 & 1 & 3 \end{pmatrix}$$

is composed by the vectors of $\mathscr{B}$ as columns. Then we compute $B^{-1}$. We have $\det(B) = -3$, whereas the cofactor matrix is given by

$$\mathrm{Cof}(B) = \begin{pmatrix} -3 & 0 & 1 \\ 0 & 3 & -1 \\ 0 & -3 & 0 \end{pmatrix}$$

Then we have

$$B^{-1} = -\frac{1}{3}\begin{pmatrix} -3 & 0 & 0 \\ 0 & 3 & -3 \\ 1 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ -\frac{1}{3} & \frac{1}{3} & 0 \end{pmatrix}$$

Then we can conclude:

$$B^{-1}AB = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ -\frac{1}{3} & \frac{1}{3} & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 3 & 5 & 8 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 3 \\ 1 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ -\frac{1}{3} & \frac{1}{3} & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 & 0 \\ 2 & 1 & 6 \\ 16 & 8 & 39 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 0 \\ 14 & 7 & 33 \\ 0 & 0 & 2 \end{pmatrix}$$

*Point 3.* We have

$$Ker(T) = \left\{ \boldsymbol{v} \in \mathbb{R}^3 \,\middle|\, T(\boldsymbol{v}) = \mathbf{0} \right\}.$$

To compute the dimension and a basis for $Ker(T)$, we have to solve the system $T\boldsymbol{v}$. For $\boldsymbol{v} = (x, y, z)$, this means

$$\begin{cases} x + z = 0\,, \\ y + z = 0\,, \\ 3x + 5y + 8z = 0\,, \end{cases} \implies \begin{cases} x = -z\,, \\ y = -z\,, \\ -3z - 5z + 8z = 0\,, \end{cases} \implies \begin{cases} x = -z\,, \\ y = -z\,. \end{cases}$$

Hence, $Ker(T) = \{(-z, -z, z) \,|\, z \in \mathbb{R}\}$. Of course, this set is generated, for example, by $\boldsymbol{v} = (-1, -1, 1)$ (it suffices to choose for $z$ any non-zero value). This means that $\dim(Ker(T)) = 1$ and a basis for $Ker(T)$ is given by $\{(-1, -1, 1)\}$.

As regards the Image, we know that it is generated by $T(\boldsymbol{e}_1)$, $T(\boldsymbol{e}_2)$, $T(\boldsymbol{e}_3)$, i.e. $(1, 0, 3)$, $(0, 1, 5)$, $(1, 1, 8)$. The dimension of $Im(T)$ is equal to $rk(A)$.

Since $\det(A) = 0$, we know that $\dim(Im(T)) < 3$. To prove that $\dim(Im(T)) = 2$, we can use the dimension theorem, who states that

$$\dim(Ker(T)) + \dim(Im(T)) = \dim \mathbb{R}^3 = 3 \implies \dim(Im(T)) = 3 - \dim(Ker(T)) = 2\,.$$

We can also observe that the submatrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

obtained from the first two columns of $A$ (hence from $T(\boldsymbol{e}_1)$ and $T(\boldsymbol{e}_2)$), has determinant different from zero (it is 1). This means that $\dim(Im(T)) = 2$ and a basis is given by

$$\{T(\boldsymbol{e}_1), T(\boldsymbol{e}_2)\} = \{(1, 0, 3), (0, 1, 5)\}\,.$$